

B Hyperparameters

We include a table of important hyperparameters below. We performed sweeps of the learning rate and batch size by an order of magnitude in both directions from the values in the table and were unable to find any significant improvements. Other reasonable choices for both the verifier temperature (eg: 1.0 instead of 0.7) and objective (cross-entropy instead of mean squared error) also had negligible effect in our ablations.

General Hyperparameters	Value
Batch Size	3.2×10^4 tokens
Max Sample Length	400 tokens
Tokenization	reversible_50000
Optimizer	Adam, $\beta_1 = 0.9$, $\beta_2 = 0.95$
Dropout	0.0
Learning Rate Schedule	Linear decay to 0
Finetuning Hyperparameters	Value
Epochs	20
Sampling Temperature	0 (argmax)
Base Learning Rate (α)	1.6×10^{-5} (3B)
	1.2×10^{-5} (6B)
	1.0×10^{-5} (12B)
	6.0×10^{-6} (175B)
Learning Rate	$0.1 \times \alpha$
Verification Hyperparameters	Value
Epochs	2 for generator, 1 for verifier
Sampling Temperature	0.7
Learning Rate	1.0×10^{-5}
Loss weight	1.0
Verifier loss	MSE
Completions per train problem	100
Completions per test problem	100

Table 1: Hyperparameters used for all experiments, unless explicitly said otherwise. Notable exceptions include Figure 8c, which uses 4x more tokens per batch and 300 completions at both training and test time. All dropout experiments in Figure 8 use 20% dropout. Figure 7a uses verifiers trained on 100 completions, but searching over more completions at test time.

C Calculator Annotations

The calculator annotations were not provided by human contractors: they were generated by a combination of hard-coded logic and a finetuned language model. The logic for auto-generating calculator annotations is imperfect. It is highly unlikely to generate any incorrect annotations, but it is not uncommon for it to ignore some lines that could be annotated.

During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens. During testing, we override model sampling when a well-formatted annotation exists, specifically overwriting the token(s) directly following “=” and within `<<...>>`.

To simulate the calculator, we simply use the python `eval` function to evaluate the tokens in the expression (Figure 9). Evaluations that time out or throw an error result in the annotations being skipped and the model being sampled from as usual.

We note that the original version of our calculator, used for all results in this paper, had some minor implementation bugs. Our reported test performance is therefore a slight underestimate, though the magnitude of this discrepancy is less than 1% in most experiments. Fixing the calculator improves verification test performance by about 1% when using the full GSM8K training set.

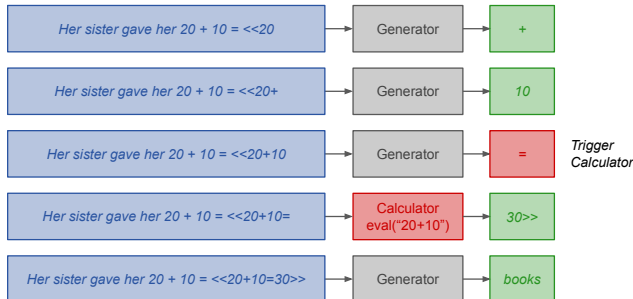


Figure 9: A diagram of the calculator sampling procedure.

D Example Model Solutions

We showcase a handful of samples comparing finetuning and verification at both 6B and 175B scale. Samples were slightly cherry-picked for diversity.

Dean's mother gave him \$28 to go to the toy store. Dean bought 6 toy cars and 5 teddy bears. Each toy car cost \$2 and each teddy bear cost \$1. His mother then feels generous and decides to give him an extra \$10. How much money does Dean have left? Ground Truth: The cost of the toy cars is 6 cars * \$2/car = \$<<6*2=12>>12. The cost of the teddy bears is 5 bears * \$1/bear = \$<<5*1=5>>5. The total cost of the toys is \$12 + \$5 = \$<<12+5=17>>17. Adding the two amounts of money his mother gave him, we find that Dean has \$28 + \$10 = \$<<28+10=38>>38 to spend. Dean has \$38 - \$17 = \$21 left. A: 21 6B Finetuning: [incorrect] The cost of the toy cars is 6 * \$2 = \$<<6*2=12>>12. The cost of the teddy bears is 5 * \$1 = \$<<5*1=5>>5. The total cost of the toys is \$12 + \$5 = \$<<12+5=17>>17. Dean has \$28 - \$17 = \$<<28-17=11>>11 left. A: 11 6B Verification: [incorrect] The cost of the toys for Dean was 6 * \$2 = \$<<6*2=12>>12. The cost of the toys for his mother was 5 * \$1 = \$<<5*1=5>>5. The total cost of the toys for both of them is \$12 + \$5 = \$<<12+5=17>>17. Dean's mother gives him an extra \$10, which makes the total amount of money he has left \$28 - \$17 + \$<<28-17>11>>11. A: 11 175B Finetuning: [correct] The cost of the toy cars is 6 * \$2 = \$<<6*2=12>>12. The cost of the teddy bears is 5 * \$1 = \$<<5*1=5>>5. The total cost of the toys is \$12 + \$5 = \$<<12+5=17>>17. Dean has \$28 - \$17 = \$<<28-17=11>>11 left. His mother gives him an extra \$10, which brings him to \$11 + \$10 = \$<<11+10=21>>21. A: 21 175B Verification: [correct] Dean spent 6*2 = \$<<6*2=12>>12 on toy cars. Dean spent 5*1 = \$<<5*1=5>>5 on teddy bears. He had 28-12-5 = \$<<28-12-5=11>>11 left. His mother gave him an extra \$10 making his total 11+10 = \$<<11+10=21>>21. A: 21	Shiela is required to submit a 15-page research paper. She already finished writing 1/3 of the paper. How many pages does she have left to write? Ground Truth: Shiela has already written one third of the paper which is 15 pages x 1/3 = <<15*1/3=5>>5 pages. So, she still needs to write 15 pages - 5 pages = <<15-5=10>>10 pages. A: 10 6B Finetuning: [correct] She has written 1/3 * 15 pages = <<1/3*15=5.0>>5 pages. The rest of the paper is 15 pages - 5 pages = <<15-5=10.0>>10 pages. A: 10 6B Verification: [correct] Shiela has written 1/3 x 15 = <<1/3*15=5.0>>5.0 pages. So, she has 15 - 5 = <<15-5=10.0>>10.0 pages left to write. A: 10 175B Finetuning: [correct] Shiela has written 1/3 x 15 = <<1/3*15=5.0>>5 pages. Therefore, she has 15 - 5 = <<15-5=10>>10 pages left to write. A: 10 175B Verification: [correct] Shiela has finished writing 1/3 x 15 = <<1/3*15=5.0>>5 pages. Thus, she has 15 - 5 = <<15-5=10.0>>10 pages left to write. A: 10 The cheese pizza is cut into 12 slices and the pepperoni pizza is cut into 8 slices. If Kate's 6 friends each eat 6 cheese pizza slices and 4 pepperoni pizza slices, how many pizza pies does she need to buy? Ground Truth: Kate's friends eat 6 x 6 = <<6*6=36>>36 cheese pizza slices. Her friends eat 6 x 4 = <<6*4=24>>24 pepperoni pizza slices. Kate needs to buy 36 / 12 = <<36/12=3>>3 cheese pizza pies. She needs to buy 24 / 8 = <<24/8=3>>3 pepperoni pizza pies. She needs to buy a total of 3 + 3 = <<3+3=6>>6 pizza pies. A: 6 6B Finetuning: [incorrect] Kate's 6 friends eat a total of 6 friends * 6 slices = <<6*6=36>>36 slices. This is a total of 36 slices + 8 slices = <<36+8=44>>44 slices needed to buy. Therefore, Kate needs 44 slices / 12 slices/pizza = <<44/12=3.6666666666666665>>3 pizza pies. A: 3 6B Verification: [incorrect] If Kate's 6 friends eat 6 cheese pizza slices each, then she needs 6*12 = <<6*12=72>>72 cheese pizza slices. If Kate's 6 friends eat 4 pepperoni pizza slices each, then she needs 4*8 = <<4*8=32>>32 pepperoni pizza slices. In total, Kate needs 72+32 = <<72+32=104>>104 pizza slices. A: 104 175B Finetuning: [incorrect] Friends eating cheese pizza will eat 12 * 6 = <<12*6=72>>72 cheese pizza slices. Friends eating pepperoni pizza will eat 8 * 4 = <<8*4=32>>32 pepperoni pizza slices. Total slices eaten will be 72 + 32 = <<72+32=104>>104 slices. Kate will need to buy 104 / 12 = <<104/12=8.666666666666667>>8 cheese pizza pies. Kate will need to buy 104 / 8 = <<104/8=13.0>>13 pepperoni pizza pies. She will need to buy 13 + 8 = <<13+8=21>>21 pizza pies. A: 21 175B Verification: [incorrect] She has 6 friends who eat 12 cheese slices each so they eat 6*12 = <<6*12=72>>72 cheese slices She has 4 friends who eat 8 pepperoni slices each so they eat 4*8 = <<4*8=32>>32 pepperoni slices Together her friends eat 72+32 = <<72+32=104>>104 slices There are 12 cheese slices in 1 pizza pie and she needs to buy 72 cheese slices so she needs 72/12 = <<72/12=6.0>>6 pizza pies There are 8 pepperoni slices in 1 pizza pie and she needs to buy 32 pepperoni slices so she needs 32/8 = <<32/8=4.0>>4 pizza pies In total she needs to buy 6 pizza pies for cheese and 4 pizza pies for pepperoni so she needs to buy 6+4 = <<6+4=10>>10 pizza pies A: 10
A perfume company is trying to create new scents. They already have 4 vanilla scents and 8 fruity scents available and they need to decide which kind of scent to focus on. They decide to focus on whichever scent sells the most and monitor their number of sales as part of their research. By the end of the day, they sell 5 of each of the vanilla scents and 2 of each of the fruity scents available. How many more vanilla scents sold compared with the fruity scents? Ground Truth: The store sold 4 types of vanilla scents * 5 sales each = <<4*5=20>>20 of the vanilla scents. The store sold 8 types of fruity scents * 2 sales each = <<8*2=16>>16 of the fruity scents. This means that the store sold 20 vanilla - 16 fruity = <<20-16=4>>4 more vanilla scents. A: 4 6B Finetuning: [incorrect] The perfume company sold 5 Vanilla A and 2 Vanilla B, so 5 + 2 = <<5+2=7>>7 Vanilla scents sold. They also sold 2 Fruity A and 5 Fruity B, so 2 + 5 = <<2+5=7>>7 Fruity scents sold. The Vanilla scents sold 7 - 5 = <<7-5=2>>2 more than the Fruity scents. A: 2 6B Verification: [correct] There were 5 vanilla scents sold in total, so they sold 5 x 4 = <<5*4=20>>20 vanilla scents. They sold 2 fruity scents in total, so they sold 2 x 8 = <<2*8=16>>16 fruity scents. Therefore, the vanilla scents sold 20 - 16 = <<20-16=4>>4 more than the fruity scents. A: 4 175B Finetuning: [correct] They sold 5 * 4 = <<5*4=20>>20 vanilla scents. They sold 2 * 8 = <<2*8=16>>16 fruity scents. They sold 20 - 16 = <<20-16=4>>4 more vanilla scents. A: 4 175B Verification: [correct] The perfume company sold 5 of each of the vanilla scents for a total of 5*4 = <<5*4=20>>20 vanilla scents. They also sold 2 of each of the fruity scents for a total of 2*8 = <<2*8=16>>16 fruity scents. Therefore, the perfume company sold 20-16 = <<20-16=4>>4 more vanilla scents than fruity scents. A: 4	